

of organisms. Similarly, all of the **biotic**, or living, factors—the other organisms that are part of an individual’s environment—also influence the distribution and abundance of life on Earth.

Global Climate Change

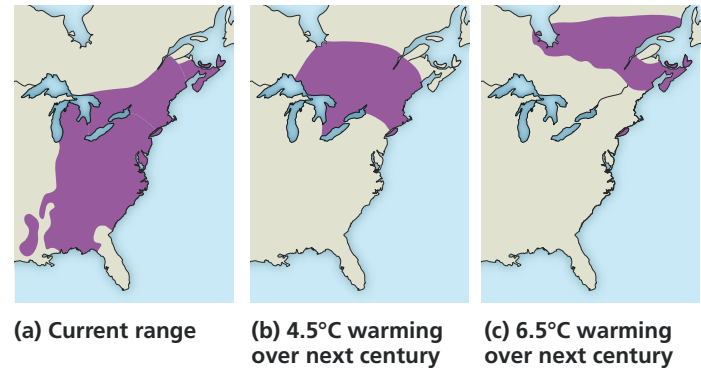
Because climatic variables affect the geographic ranges of most plants and animals, any large-scale change in Earth’s climate profoundly affects the biosphere. In fact, a large-scale climate “experiment” is under way: The burning of fossil fuels and deforestation are increasing the concentrations of carbon dioxide and other greenhouse gases in the atmosphere. This has caused **climate change**, a directional change to the global climate that lasts three decades or more (as opposed to short-term changes in the weather). As we’ll explore in more detail in Concept 56.4, Earth has warmed an average of 0.9°C (1.6°F) since 1900 and is projected to warm 1–6°C (2–11°F) more by the year 2100. Wind and precipitation patterns also are shifting, and extreme weather events (such as major storms and droughts) are occurring more frequently.

How will such changes affect the distribution of organisms? One way to answer this question is to look at the changes that have occurred since the last ice age ended. Until about 16,000 years ago, glaciers covered much of North America and Eurasia. As the climate warmed and the glaciers retreated, tree species distributions expanded northward. Fossil pollen shows that while some species moved northward rapidly, others moved more slowly, with their range expansion lagging several thousand years behind the shift in suitable habitat.

Will plants and other species be able to keep up with the much more rapid warming projected for this century? Consider the American beech, *Fagus grandifolia*. Ecological models predict that the northern limit of the beech’s range may move 700–900 km northward in the next century, and its southern range limit will shift even more. The current and predicted geographic ranges of this species under two different climate-change scenarios are illustrated in **Figure 52.8**. If these predictions are even approximately correct, the beech’s range must shift 7–9 km northward per year to keep pace with the warming climate. However, since the end of the last ice age, the beech has moved at a rate of only 0.2 km per year. Without human help in moving to new habitats, species such as the American beech may have much smaller ranges or even become extinct.

Indeed, the climate change that has *already* occurred has affected the geographic ranges of hundreds of terrestrial, marine, and freshwater organisms. For example, as the climate has warmed, 22 of 35 European butterfly species studied have shifted their ranges farther north by 35–240 km in recent decades. In western North America, nearly 200 plant species have moved to lower elevations, most likely in response to decreased rain and snow at higher elevations. Other research shows that a Pacific diatom species, *Neodenticula seminae*, recently colonized the Atlantic Ocean for the first time in 800,000 years. In these and many other

▼ **Figure 52.8** Current range and predicted ranges for the American beech under two climate-change scenarios.



? The predicted range in each scenario is based on climate factors alone. What other factors might alter the distribution of this species?



Mastering Biology

Interview with Margaret Davis: Using fossil pollen to track tree species migration



such cases, when climate change enables or causes a species to move into a new geographic area, other organisms living there may be harmed (see Figure 56.31).

Furthermore, as the climate changes, some species are facing a shortage of suitable habitat while others cannot migrate quickly enough. For example, a recent study found that on average, the geographic ranges of 67 bumblebee species (**Figure 52.9**) in the Northern Hemisphere were shrinking: They were retreating from the southern edges of their distributions, but failing to expand their ranges to the north. Overall, climate change is causing the populations of many species to decrease in size or even disappear (see Figure 1.12).

▼ **Figure 52.9** The rusty-patched bumblebee (*Bombus affinis*).



CONCEPT CHECK 52.1

1. Explain how the sun’s unequal heating of Earth’s surface results in deserts near 30° north and south of the equator.
2. What are some of the differences in microclimate between an unplanted agricultural field and a nearby stream corridor with trees?
3. **WHAT IF?** Changes in Earth’s climate at the end of the last ice age happened gradually, taking centuries to thousands of years. If the planet continues to warm at its current, rapid rate, how might this affect the evolution of long-lived trees compared with that of annual plants, which have much shorter generation times?
4. **MAKE CONNECTIONS** Focusing just on the effects of temperature, would you expect the global distribution of C₄ plants to expand or contract as Earth becomes warmer? Why? (See Concept 10.4.)

For suggested answers, see Appendix A.